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SECONDARY MAGNETIZATION OF TRIASSIC-JURASSIC VOLCANICLASTIC ROCKS OF THE QUESNEL TERRANE, QUESNEL LAKE, B.C.

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Abstract. Upper Triassic-Lower Jurassic rocks at Quesnel Lake form part of the Quesnel terrane, which was emplaced against the Omineca Belt, part of the North American (paleo-)continental margin, in the late Early Jurassic. Hard magnetizations, which may or may not be primary, were identified in volcanic wackes, but they are poorly grouped. However, removed magnetizations of intermediate unblocking temperatures (200-500°C) and coercivities (30-50 mT) are systematically directed northeast and downward (D=35°, I=57°, a_{95}=9°, 5 sites, 22 specimens; paleopole 62°N, 15°W, dm=13°, dp=10°) with respect to present horizontal. We interpret this to be an overprint, acquired after Middle to Late Jurassic folding, as a result of mid-Cretaceous plutonism during the Cretaceous Normal Superchron. The paleopole is compatible with Early to mid-Cretaceous paleopoles from other terranes in the southern Canadian Cordillera, but does not agree with observations from post-Jurassic cratonic rocks. This disagreement indicates that in the mid-Cretaceous these terranes, along with at least part of the Omineca Belt to which they were attached, formed a quasi-rigid super-terrane which was situated about 2000 km farther south than its present position relative to the craton.

Introduction

In what is now the southern Canadian Cordillera, the first oceanic terrane to be emplaced against the ancient continental margin of western North America (now the Omineca Belt) in the early Mesozoic was Terrane I (Figure 1; Monger et al., 1982; Monger, 1984). Terrane I is a composite of several smaller terranes, one of which is the Quesnel terrane. Paleomagnetic results from the Quesnel terrane are described.

The Quesnel terrane, as part of Terrane I, was thrust onto the Omineca Belt in the late Early Jurassic, and together they were deformed, metamorphosed, and intruded by syn- to post-tectonic plutons in the Middle to Late Jurassic (Read and Wheeler, 1976; Archibald et al., 1983; Brown et al., 1985). It is important to note that the rocks in the sampling locality are structurally continuous with the adjacent Omineca Belt, indicating that since the Quesnel terrane was emplaced, there has been no significant relative movement between them.

Geology

At Quesnel Lake (Figure 2; 52.5°N, 121.5°W) zeolite facies rocks approximately equivalent to the Upper Triassic-Lower Jurassic Nicola Group and Takla Group, comprise mainly alkali basalt lavas and volcaniclastic rocks, and comagmatic Early Jurassic plutons (Morton, 1976). Small intrusions of Cretaceous age are widespread in the area, such as the Bootjack Stock with a whole-rock K-Ar age of 111 ± 4 Ma (Bailey, 1978). Collections were made in bedded green-grey volcanic wackes, siltstones and mudflow deposits, which are dated by fossils as Norian and Hettangian. Beds dip up to about 70°. Because of the continuity with structures in the Omineca Belt, this folding is correlated with the regional Middle to Late Jurassic deformation. There is no evidence for major post-Jurassic folding.

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Method

Oriented hand samples were collected at nine sites and cored in the laboratory. Ninety-three specimens (64 cores) were then step-demagnetized by heating or alternating fields (AF). The directions of the untreated natural remanent magnetizations (NRM) are widely scattered. Detailed demagnetization up to 700°C and 100 mT was carried out on pilot specimens from each site. The remaining specimens were then demagnetized in somewhat less detail. Constituent magnetizations were identified by linear segments on orthogonal plots (Zijderveld, 1967) or by stable endpoints (McElhinny and Gough, 1963).

Results

Only 59 specimens yield resolvable magnetizations. At site 7 directions are close to the geocentric axial dipole field and remained unchanged during demagnetization. We regard this as low temperature viscous remanent magnetization acquired during the Bruhnes chron. Other specimens generally have two magnetizations and examples are given in Figure 3. During thermal demagnetization, the direction of a specimen from site 10 migrates along a great circle between 200 and 500°C, corresponding to a linear segment that does not decay to the origin (Figure 3a). The removed magnetization (RM) is directed to the northeast and downward with respect to present horizontal. A very small stable magnetization with unblocking temperatures in excess of 600°C is also present. The intensity decreases steadily during treatment.

Thermal demagnetization of a specimen from site 8 records a migration between 100 and 500°C corresponding to a linear trend on the orthogonal plot, and a northeasterly, downward directed removed magnetization (RM, Figure 3b). Again, a stable magnetization with unblocking temperatures in excess of 600°C is present, and gives a recognizable linear trend.

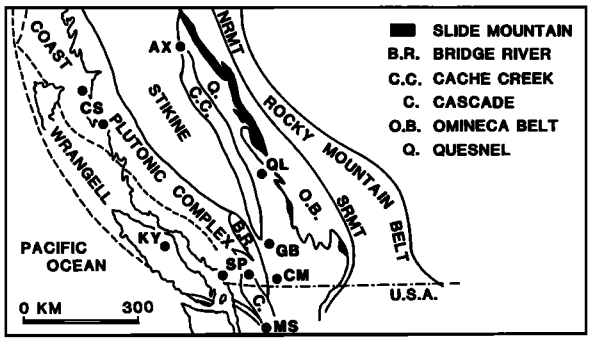


Fig. 1. Tectonic belts and terranes of the southern Canadian Cordillera and northern U.S., from Monger et al. (1982) and Monger (1984). NRMT, SRMT, Northern and Southern Rocky Mountain Trench. Stikine, Cache Creek, Quesnel and Slide Mountain terranes form Terrane I. Paleomagnetism localities: QL, Quesnel Lake; AX, Axelgold intrusion; GB, Guichon batholith; CM, Copper Mountain intrusions; CS, Captain Cove-Stephens Is. plutons; SP, Spuzzum-Porteau plutons; MS, Mount Stuart batholith; KY, Karmutsen Formation.

AF demagnetization gives less definitive results, but scattered southerly or easterly, upward directions are usually obtained after treatment at 50 mT, and the magnetization removed is generally directed northeast and downward (Figure 3c).

The directions of the high stability magnetizations (T >600°C, AF >50 mT) show no general coherence over the rock-unit as a whole. They are older than the Middle to Late Jurassic deformation because the site means are better grouped with respect to bedding. They could be primary magnetizations acquired during deposition or Early Jurassic plutonism, but their origin is uncertain and we make no attempt to interpret them.

In contrast, the removed magnetizations, which we refer to collectively as QL, have intermediate unblocking temperatures (200-500°C) and coercivities (30-50 mT), and are systematically directed northeast and downward (mean direction, D=34.9°, I=57.4°, k=12, a_{95}=9°, 5 sites, 22 specimens, Figure 4a; paleopole, 61.9°N, 14.6°W, dm=13.4°, dp=9.8°). None are upwardly directed. QL was acquired after Jurassic deformation because the specimen (and site means) directions are more poorly grouped (k=2) with respect to bedding.

Interpretation

We interpret the QL magnetization to be a post-Jurassic overprint. The directions of the present earth's field and the geocentric axial dipole field both lie outside the a_{95} circles of confidence (Figure 4a). Hence, the magnetization is not recent.

The paleopole is not compatible with any part of the Mesozoic or Cenozoic APW path for cratonic North America (Figure 4b). However, it is situated among the paleopoles obtained from Early to mid-Cretaceous intrusions from the Cache Creek terrane (AX, Irving et al., 1985), the Cascade terrane (MS, Beck et al., 1981) and the Coast Plutonic Complex (SP, Irving et al., 1985, and CS, Symons, 1977). The Coast Plutonic Complex is a magmatic arc marking the joining of a second composite terrane, Terrane II of Monger et al. (1982) (Wrangell in Figure 1), to Terrane I in the mid-Cretaceous. Thus, geological evidence that Terranes I and II were together by the mid-Cretaceous, is supported paleomagnetically by the fact that paleopoles of this age from both terranes fall in the same general region. The QL magnetization is interpreted to be also mid-Cretaceous because its paleopole falls in this group, and because no reversals are observed, consistent with an origin during the Cretaceous Normal Superchron (118 to 83 Ma; Harland et al., 1982). Further support is derived from the fact that the linear form of the thermal decay curves (Figure 3a) is typical of viscous partial thermoremanence (PTRM, Irving and Opdyke, 1965), which may be reasonably attributed to heating related to the Cretaceous plutonic activity in the region. This activity was evidently also responsible for resetting biotite K-Ar ages: a nearby Early Jurassic intrusion was reset to 105 ± 4 Ma (White et al., 1968), and a Paleozoic intrusion was reset to 108 ± 4 Ma (Stevens et al., 1982, p.13).

Other paleopoles lie within this Cretaceous group. KY is a secondary magnetization from the Upper Triassic Karmutsen Formation of Vancouver Island (part of Terrane II) which could be a chemical or viscous PTRM overprint at least partly of Cretaceous age (Yole and Irving, 1980). CM and GB are paleopoles from the Early Jurassic Copper Mountain intrusions and the Guichon Batholith (Symons, 1971, 1983, 1984), both from the southern part of the Quesnel terrane. Symons interpreted them as primary, Early Jurassic magnetizations, but we suggest that they could be Cretaceous thermal overprints, on the basis of their paleopoles. Plutonism of this age is widespread in southern British Columbia and could have provided the necessary regional heating.

The QL magnetization indicates a paleolatitude of 38 ± 10°N (Figure 4c). The expected paleolatitude, with reference to the cratonic pole (NA, Figure 4b), taken at 67.7°N, 186.3°E (A_{95}=3.8°) for 94 Ma (Irving et al., 1985), is 61 ± 4°N, indicating a relative latitudinal displacement of 23 ± 10° to the north. The

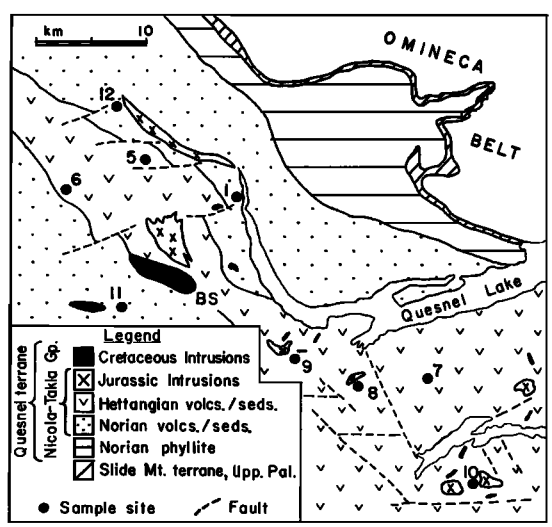


Fig. 2. Simplified geological map of the sampling area. BS, Bootjack Stock.

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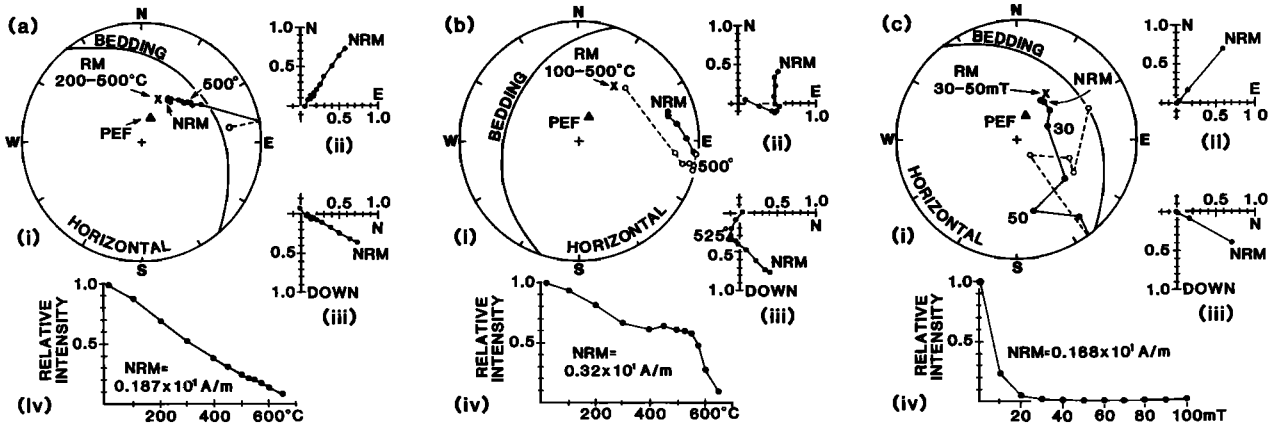


Fig. 3. (a) Thermal demagnetization in site 10 and (b) site 8. (c) AF demagnetization in site 10. In each: (i) direction changes (solid circles, downward; open circles, upward), (ii) and (iii) orthogonal plots, (iv) intensity decay. PEF, present earth's field; RM, removed magnetization (X). Note: orthogonal plots are with respect to paleohorizontal; directions differ from stereograms.

observed rotation relative to the cratonic paleonorth is 73 ± 15° clockwise.

This result is very similar to those for the other magnetizations mentioned above, which are also normally polarized (Figure 4c; see Irving et al., 1985). Taken together, they imply that Terranes I and II were together in the mid-Cretaceous, as indicated by geological evidence (see Monger, 1984), and that between the mid-Cretaceous and the mid-Tertiary they have moved northwards relative to the craton by roughly 2000 to 2500 km, and rotated clockwise by about 70°, as a quasi-rigid super-terrane. At least the western part of the southern Omineca Belt was part of this super-terrane because it was attached to Terrane I by the Middle Jurassic. The eastern limit of the super-terrane is presently not defined.

This conclusion is based on paleomagnetic data, a reasoned interpretation of age of magnetization, and the timing of terrane amalgamation from geological evidence. It is not a tectonic model. Any proposed model for this super-terrane and its relationship with the cratonic margin must somehow reconcile their paleomagnetic discordance with the compelling geological links between them. Notwithstanding Chamberlain and Lambert's (1985) attempt to do this, more work must be done before the enigma is solved. For example, because North America was at its most northerly position in the mid-Cretaceous, the paleomagnetic disparity with the super-terrane and implicit relative movement are at a maximum. A revised, younger or older age for QL and the other magnetizations would reduce the displacement, perhaps into the range of the roughly 1000 km of Late Cretaceous through ?Eocene dextral strike-slip fault movement documented in the Tintina-Northern Rocky Mountain Trench system (Gabrielse, 1985). However, the

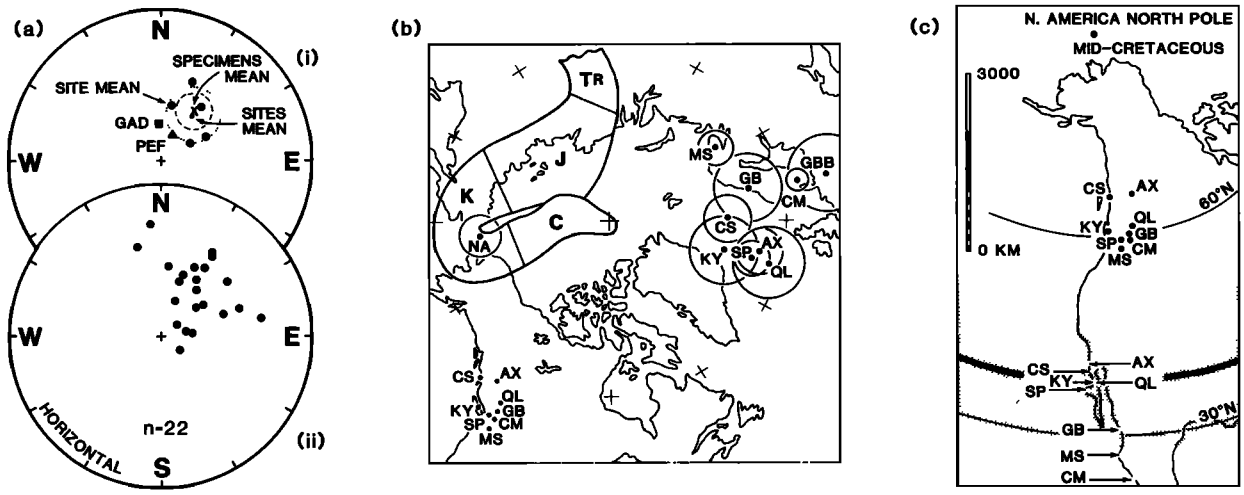


Fig. 4. (a) Removed magnetizations. (i) Site means; specimen mean, a_{95} dashed; sites mean, a_{95} dotted. GAD, geocentric axial dipole field. (ii) Individual specimens. (b) Paleopoles of localities in Figure 1, and standard error circles (P=0.37); North America APW path and 100-80 Ma paleopole (NA). GB, Symons (1971); GBB, Symons (1983). All paleopoles have been calculated from directions of magnetization relative to present horizontal. Absence of tilting is assumed. (c) Paleolatitudes of mid-Cretaceous or presumed mid-Cretaceous magnetizations from B.C. and northern U.S., with reference to North America for 100-80 Ma. Longitude is indeterminate. Paleolatitude and error band shown for QL.

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interpreted mid-Cretaceous ages are certainly the most reasonable at present.

An alternative explanation to latitudinal displacements is a tilting of the mid-Cretaceous paleohorizontal of about 30° to the southwest, which would have the effect of rotating the expected paleofield directions to the observed directions. This possibility must be investigated, but such a consistent tilt over much of British Columbia and northern Washington is unlikely. Tilting may account for part of the paleomagnetic anomaly, but probably not for all of it. More likely, tilt was non-uniform, producing a second order effect such as the dispersion of the paleopoles.

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